Frame Partitioning Algorithms

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Abstract

In the field of signal reconstruction, data loss problems are addressed using a special class of vector sets called **finite frames**, which are generalized bases with built-in redundancy. This project seeks to find partitioning algorithms of these finite frames such that a particular matrix norm bound is satisfied. We utilized analytic and algebraic techniques to understand the eigenvalues of matrices that are formed from two special classes of finite frames—Equiangular frames and Grassmannian frames. Our research has uncovered algorithms to generate partitions of these frames given certain conditions, and some frames can be trivially partitioned compared to other frames. Our analysis is ongoing with the goal of finding efficient partitioning algorithms.

Preliminaries

Eigenvalues & Eigenvectors

For an $n \times n$ matrix $\mathbf{M}$, scalars λ and nonzero vectors $v \in \mathbb{C}^n$ satisfying $\mathbf{M}v = \lambda v$ are called *eigenvalues* and *eigenvectors* of $\mathbf{M}$. The eigenvalues are obtained by solving the *characteristic* equation

$$\det(\mathbf{M} - \lambda \mathbf{I}) = 0.$$

Induced Matrix Norm (Spectral Norm)

A vector norm that is defined on $\mathbb{C}^n$ *induces* a matrix norm on $\mathbb{C}^{n\times n}$ by setting

$$\|\mathbf{M}\| = \max_{\|x\|=1} \|\mathbf{M}x\| = \max\{\lambda_1, \lambda_2, ...\}.$$

Where $\mathbf{M}$ is a **square** matrix in $\mathbb{C}^{n\times n}$, $x \in \mathbb{C}^n$, and $\lambda_1, \lambda_2, ...$ are the eigenvalues of $\mathbf{M}$.

Tensor Product

Given two vectors $x, y \in \mathbb{C}^n$ the tensor product (outer product) of two vectors is defined as

$$x \otimes y = xy^* = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} \overline{y}_1 & \overline{y}_2 & \cdots & \overline{y}_n \end{bmatrix} = \begin{bmatrix} x_1\overline{y}_1 & x_1\overline{y}_2 & \cdots & x_1\overline{y}_n \\ x_2\overline{y}_1 & x_2\overline{y}_2 & \cdots & x_2\overline{y}_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n\overline{y}_1 & x_n\overline{y}_2 & \cdots & x_n\overline{y}_n \end{bmatrix}$$

Frames

Definition: Finite Frames

A **frame** for a complex space $\mathbb{C}^n$ is a sequence of vectors $\mathcal{F} = \{f_i\} \subset \mathbb{C}^n$ for which there exist constants $0 < A \leqslant B < \infty$ such that, for every $x \in \mathbb{C}^n$,

$$A\|x\|^2 \leqslant \sum_i |\langle x, f_i\rangle|^2 \leqslant B\|x\|^2$$

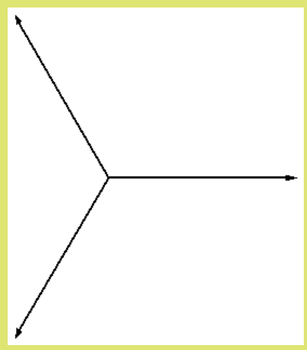
- A, B are called frame bounds.
- The set of frame vectors $\mathcal{F}$ spans $\mathbb{C}^n$.

Equiangular-Equinorm Frames

Let our frame $\mathcal{F} =$

$$\{f_j\}_{j=0}^{m-1} = \sqrt{\frac{2}{m}} \begin{bmatrix} \cos\left(\frac{2\pi j}{m}\right) \\ \sin\left(\frac{2\pi j}{m}\right) \end{bmatrix}_{j=0}^{m-1}$$

Mercedes-Benz Frame



Example: Mercedes-Benz Frame

Let $m = 3$ and we have $\mathcal{F} = \{f_0, f_1, f_2\} =$

$$\left\{ \begin{bmatrix} \sqrt{\frac{2}{3}} \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}.$$

Maximal Frame Correlation & Grassmannian Frames

For a given **unit norm frame** $\mathcal{F} = \{f_k\}_{k=1}^n$ in $\mathbb{E}^m$ we define the maximal frame correlation $\mathcal{M}(\mathcal{F})$ by

$$\mathcal{M}(\mathcal{F}) = \max_{k,l,k\neq l} \{|\langle f_k, f_l\rangle|\}$$

A frame that **minimizes** $\mathcal{M}$ is called **Grassmannian**.

Grassmannian Construction

To construct a Grassmannian frame we choose a value for m.

1. Recursively find the $2m \times 2m$ conference matrix $\mathbf{C}$. This is a matrix of ± 1's and the main diagonal is all 0's, and $\mathbf{C}\mathbf{C}^T = (n-1)\mathbf{I}$.
2. Calculate the $\mathbf{R}$ matrix: $\mathbf{R} = \frac{i}{\sqrt{2m-1}}\mathbf{C} + \mathbf{I}$.
3. Find the factorization $\mathbf{R} = \mathbf{W}\Lambda\mathbf{W}^*$ and the columns of $\mathbf{W}$ are eigenvectors.
4. Take the *left* half rows of $\mathbf{W}$, these are the frame vectors.

The MSS Theorem

Let r be a positive integer and $u_1, ..., u_m \in \mathbb{C}^d$ be vectors such that

$$\sum_{i=1}^m u_i \otimes u_i = \mathbf{I}$$

and $\|u_i\|^2 \leqslant \delta$ for all i. Then there exists a **partition** $\{S_1, ..., S_r\}$ of $[m]$ such that

$$\left\| \sum_{i\in S_j} u_i \otimes u_i \right\| \leqslant \left(\frac{1}{\sqrt{r}} + \sqrt{\delta} \right)^2 \qquad \text{(MSS*)}$$

for $j = 1, ..., r$.

Application of MSS to Frames

In the context of our research we utilize frames $\mathcal{F} = \{f_1, f_2, ..., f_m\}$ that fulfill the hypothesis of the MSS theorem.

RESULTS: Equinorm-Equiangular Frames

For Equinorm-Equiangular Frames the following partitions are valid:

- *Even* split partition,
- Unequal split partition,
- and *almost* even split partition

Example: Mercedes-Benz Frame

Suppose we have the following three vectors (these form the *Mercedes-Benz* frame)

$$\mathcal{F} = \{f_1, f_2, f_3\} = \left\{ \begin{bmatrix} \sqrt{\frac{2}{3}} \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}.$$

- Suppose we let $r = 2$,
- The MSS inequality bound is $\left(\frac{1}{\sqrt{r}} + \sqrt{\delta}\right)^2 = \left(\frac{1}{\sqrt{2}} + \sqrt{\frac{2}{3}}\right)^2 = \frac{7}{6} + \frac{2}{\sqrt{3}}$

Let $S_1 = \{1\}$ and $S_2 = \{2, 3\}$

$$S_1 \Rightarrow \mathcal{P}_1 = \{f_1\} = \left\{ \begin{bmatrix} \sqrt{\frac{2}{3}} \\ 0 \end{bmatrix} \right\}$$

$$S_2 \Rightarrow \mathcal{P}_2 = \{f_2, f_3\} = \left\{ \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right\}$$

The corresponding norms of the tensor product sums for $\mathcal{P}_1, \mathcal{P}_2$ are

$$\|f_1 \otimes f_1\| = \frac{2}{3} < \frac{7}{6} + \frac{2}{\sqrt{3}}$$

$$\|f_2 \otimes f_2 + f_3 \otimes f_3\| = 1 < \frac{7}{6} + \frac{2}{\sqrt{3}}$$

$\Rightarrow \{S_1, S_2\}$ is a *valid* partition of $\mathcal{F}$.

RESULTS: Grassmannian Frames

For Grassmannian Frames the following was observed:

- For $m = \{2, 4, 8\}$, *any* **partition** will be valid for the MSS inequality.
- For the Grassmannian frame $m = 16$ partitions with subsets of **consecutive** vectors are valid partitions, and for *non-consecutive* cases we obtain invalid partitions.

Example: Grassmannian Frames

The Grassmannian construction will yield $n = 2m$ frame vectors and each vector is in $\mathbb{C}^m$.
Let $m = 2 \Rightarrow n = 4$ frame vectors and

$$\mathcal{F} = \{f_1, f_2, f_3, f_4\} = \left\{ \begin{bmatrix} \frac{1}{6}(3i - \sqrt{3}) \\ \frac{i}{\sqrt{6}} \end{bmatrix}, \begin{bmatrix} \frac{1}{6}(3i + \sqrt{3}) \\ -\frac{i}{\sqrt{6}} \end{bmatrix}, \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{i}{\sqrt{6}} \end{bmatrix} \right\}$$

We need to find a **partition** of $\mathcal{F}$, this means finding subsets of $\{1, 2, 3, 4\}$.

- The left column lists every possible subset of $[4] = \{1, 2, 3, 4\}$. Notice we have $2^{2m} - 1 = 15$ possible sets, a power set.
- The right columns show the corresponding norm of the sum of tensor products.
- It turns out that each norm in the right column is less than the bound described by the MSS inequality. This means that *any* **partition** of $\{1, 2, 3, 4\}$ is valid for the MSS theorem.
- If we let $m = 16 \Rightarrow n = 32$, we have 32 vectors in $\mathbb{C}^{16}$. In this case **not every** partition of $\{1, 2, ..., 32\}$ is valid.
- As an example if we **partition** $\{1, 2, ..., 32\}$ into $r = 17$ subsets and make one subset correspond to *non-consecutive* vectors, this will form an *invalid* partition.

S	$\|\mathbf{F}_S\|$
$\{1\}$	0.5
$\{2\}$	0.5
$\{3\}$	0.5
$\{4\}$	0.5
$\{1, 2\}$	0.788675...
$\{1, 3\}$	0.788675...
$\{1, 4\}$	0.788675...
$\{2, 3\}$	0.788675...
$\{2, 4\}$	0.788675...
$\{3, 4\}$	0.788675...
$\{1, 2, 3\}$	1
$\{1, 2, 4\}$	1
$\{1, 3, 4\}$	1
$\{2, 3, 4\}$	1
$\{1, 2, 3, 4\}$	1

*Reference: A. Marcus, D. A Spielman, and N. Srivastava. *Interlacing Families II: Mixed Characteristic Polynomials and the Kadison - Singer Problem.* In: Annals of Mathematics 182 (June 2013), pp. 327 - 350

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